Questions

1. **Base Class Design**
    - Create a base class Employee with:
        - Private members: string name, int employeeID.
        - Protected member: double baseSalary.
        - Public members: Constructor, getID(), pure virtual displayInfo(), and virtual calculatePay().
        - Why must name be private, and how can derived classes access it indirectly?
2. **Intermediate Class Design**
    - Define a class SeniorEmployee that inherits publicly from Employee.
    - Add a private member double experienceBonus (based on years of experience).
    - Override calculatePay() to include baseSalary + experienceBonus.
    - Override displayInfo() to display name, ID, base salary, and experience bonus.
    - How does the protected baseSalary behave in SeniorEmployee, and can it be accessed directly?
3. **Derived Class Design**
    - Create a class DepartmentHead that inherits publicly from SeniorEmployee.
    - Add a private member double departmentBudget (a fixed amount they manage).
    - Override calculatePay() to include baseSalary + experienceBonus + 10% of departmentBudget.
    - Override displayInfo() to show all details (name, ID, base salary, experience bonus, department budget).
    - Can DepartmentHead access name directly? If not, what modification is needed in the hierarchy?
4. **Access Specifier Exploration**
    - Attempt to access baseSalary from main() using a DepartmentHead object. Why does this fail?
    - Add a protected getter getName() in Employee. Can DepartmentHead use it? Can main() use it? Explain.
5. **Multilevel Hierarchy**
    - Declare an array of Employee* pointers in main().
    - Store objects of SeniorEmployee and DepartmentHead in the array.
    - Call displayInfo() and calculatePay() for each. What output do you expect?

6. **Design Challenge**
    - Add a private member int yearsOfExperience in SeniorEmployee and calculate experienceBonus as yearsOfExperience * 1000. How would you initialize this in the constructor?

- If DepartmentHead needs to access yearsOfExperience, what changes are required in SeniorEmployee?